

JWALIN SHAH

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AI Systems Engineer · Agent Infrastructure · Evaluation · Local-First AI

Systems that constrain, observe, and verify probabilistic agents — so success is measured, not claimed.

EXPERIENCE

Core Contributor — OpenHuman (tinyhumansai)

Remote · Mar 2026 – Present

- Own **life_capture** / **curated_memory** in the OpenHuman Rust core: on-device personal index ingesting iMessage, Gmail, Calendar, and Contacts into a structured KB the agent reasons over — the memory layer for a 35k+ star local-first personal AI.
- Shipped the stack end-to-end: hybrid retrieval (SQLite FTS + sqlite-vec), ingest-before-embed correctness, contact resolution/scoring, EventKit + CNContactStore bridges, chronicle dispatcher, session management. 80+ upstream commits spanning memory, chat reliability, and security-policy hardening.

Data Operations Lead (ML Systems & Reliability) — Skild AI

San Mateo, CA · Jul – Nov 2025

- Built data validation, collection, and evaluation systems across five robotic platforms. Python automation cut overhead 50% and improved task success 40%. Hired/onboarded 30+ operators; authored training systems and runbooks that scaled the team.
- Primary technical operator for 25+ live robot demos during a \$1B Series C — zero-failure execution under investor and partner pressure.

AI Systems Engineer — Break the Web / LiveLM

Remote · 2026

- Retrieval + evaluation infra grounding LLMs in real-time knowledge with explicit hit / partial / stale / miss routing — abstain instead of invent. MCP tool-use eval harness: 8 independently measured stages; 71-question / 512-probe benchmark across serving paths.

SELECTED PROJECTS

Bridge — Reliable Coding-Agent Execution

Go · macOS Seatbelt · Worktrees

- Spawn / verify / deliver pipeline with default-deny OS sandboxing. Agents in isolated worktrees with declared paths; scope violations fail closed and re-verify in a fresh checkout — the agent cannot certify its own success. Fixed a fail-open gate; hardened 6 adapters; catalogued 9 failure modes.

Voice Engine — Local Dictation, Measured Model Selection

Swift · CoreML · ANE

- Benchmarked 26 ASR models / 560 clips. Chose moonshine-tiny (0.2177 WER, 0.05 RTF) over a near-equal 2.5B model 14× slower — real-time on device, no network. Root-caused a CoreML production failure to an OS regression with a 4-float minimal repro.

Tensor Logic — Empirical Evaluation of Domingos (2025)

Go · $\mathbb{Z}_2$ Algebra

- 3-scalar TL recurrence trained once on 16-node graphs: F1≈1.000 zero-shot to n=1024; 71M MLP retrained per size ≈0. Validated on OSS import graphs to 1,532 nodes; documented where the formalism breaks (parity/XOR).

OfficeQA Arena — Grounded Financial Reasoning (Sentient Cohort 0)

Eval · Retrieval

- 184.5/246 (75%) at \$1.71 across 9 architecture generations and 3,600+ evaluations. Showed retrieval/tool-selection errors — not model reasoning — drive most failures; shell grep beat an 11GB multi-component pipeline.

TECHNICAL SKILLS

Agent infra: Seatbelt sandboxing, worktree isolation, spawn/verify gates, MCP tool-use eval · **AI systems:** hybrid retrieval, MLX/CoreML on-device inference, hallucination & abstention measurement · **Languages:** Go, Rust, Python, Swift, TypeScript, SQLite + sqlite-vec

EDUCATION

University of Texas at Dallas — B.S. Computer Science · GPA 3.67 · Dec 2024

Ultimate Frisbee Captain · USCF 1832